Lesson 3.5

Equations of Parallel & Perpendicular Lines

Same slope, or flip and change the sign

Slope rules

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Flip the fraction **and** change the sign: $\frac{2}{3} \to -\frac{3}{2}$ $\left(\frac{2}{3} \cdot -\frac{3}{2} = -1\right)$

Find the slopes

$y = \frac{2}{3}x + 1$: slope $\frac{2}{3}$

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Parallel slope: $\frac{2}{3}$

Perpendicular slope: $-\frac{3}{2}$

Write a parallel line

Through $(2, 3)$, parallel to $y = 4x - 1$

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Same slope $m = 4$: $y - 3 = 4(x - 2)$

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Same slope $m = 4$: $y - 3 = 4(x - 2)$

$y = 4x - 5$

Write a perpendicular line

Through $(4, 1)$, perpendicular to $2x + 3y = 6$

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Solve for y: $y = -\frac{2}{3}x + 2$, slope $-\frac{2}{3}$

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Through $(4, 1)$, perpendicular to $2x + 3y = 6$

Solve for y: $y = -\frac{2}{3}x + 2$, slope $-\frac{2}{3}$

Perpendicular slope $\frac{3}{2}$: $y - 1 = \frac{3}{2}(x - 4) \Rightarrow y = \frac{3}{2}x - 5$

Watch out

Watch out! Perpendicular needs **both** steps: flip **and** change the sign. The perpendicular slope of 2 is $-\frac{1}{2}$, not $\frac{1}{2}$.

Your turn! Pause the video and try it.

Write the line through $(3, -2)$ perpendicular to $y = -3x + 7$.

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Write the line through $(3, -2)$ perpendicular to $y = -3x + 7$.

Slope $\frac{1}{3}$: $y + 2 = \frac{1}{3}(x - 3) \Rightarrow y = \frac{1}{3}x - 3$

Remember

- **Parallel**: same slope.
- **Perpendicular**: negative reciprocal — flip and change the sign.
- Then use point-slope form: $y - y_1 = m(x - x_1)$.

Now practice at your level:

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